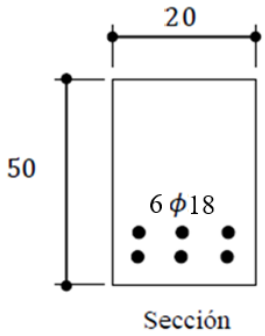
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Determine la resistencia nominal de la viga detallada en la figura. Considere hormigón G20 y acero A630-420H. Datos: $r := 5\ cm$ (al centroide de la fila inferior)



Sección

Datos

$n_b := 6$ $n_{filas} := 2$ $\phi_b := 18\ mm$ $b := 20\ cm$ $h := 50\ cm$ $r := 5\ cm$

$d_1 := h - r = 0.45\ m$ $d_2 := h - r - \phi_b - 2.5\ cm = 40.7\ cm$

$f_c := 200\ \frac{kgf}{cm^2}$ $f_y := 4200\ \frac{kgf}{cm^2}$ $\beta := 0.85$

$A_s := \pi \cdot \frac{\phi_b^2}{4} \cdot n_b = 15.2681\ cm^2$

Solución

Método aproximado

$$d_{eq} := \frac{d_1 + d_2}{2} = 42.85\ cm$$

$$\rho := \frac{A_s}{b \cdot d_{eq}} = 0.0178 \qquad \rho_b := \frac{0.85 \cdot \beta \cdot f_c}{f_y} \cdot \frac{6300\ \frac{kgf}{cm^2}}{6300\ \frac{kgf}{cm^2} + f_y} = 0.0206$$

$\rho < \rho_b$ Falla dúctil

$$M_n := \rho \cdot b \cdot d_{eq}^2 \cdot f_y \cdot \left(1 - 0.59 \cdot \rho \cdot \frac{f_y}{f_c}\right) = 21.4126\ tonnef\ m$$

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Método Exacto

$A_{s1} := \frac{A_s}{n_{filas}} = 7.6341\ cm^2$

$A_{s2} := A_{s1}$

Equilibrio de fuerzas (Suponer que A_{s1} y A_{s2} fluyen)

C = T

$0.85 \cdot f_c \cdot b \cdot a = A_{s1} \cdot f_y + A_{s2} \cdot f_y$

$a := \frac{A_{s1} \cdot f_y + A_{s2} \cdot f_y}{0.85 \cdot f_c \cdot b} = 18.8606\ cm$ $\quad c := \frac{a}{\beta} = 22.189\ cm$

$\varepsilon_c := 0.003$

$\varepsilon_{s1} := \frac{\varepsilon_c}{c} \cdot (d_1 - c) = 0.0031$ $\quad \varepsilon_{s1} > \varepsilon_y$ Fila de aceros 1 fluye ==> Se verifica el supuesto

$\varepsilon_{s2} := \frac{\varepsilon_c}{c} \cdot (d_2 - c) = 0.0025$ $\quad \varepsilon_{s2} > \varepsilon_y$ Fila de aceros 2 fluye ==> Se verifica el supuesto

$M_n := A_{s1} \cdot f_y \cdot \left(d_1 - \frac{a}{2}\right) + A_{s2} \cdot f_y \cdot \left(d_2 - \frac{a}{2}\right) = 21.4308\ tonnef\ m$

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